

6CS012 – Artificial Intelligence and Machine Learning.

Tutorial – 01

Foundational **Math Skills** for AI and ML.

A quick revision on **Derivative** for **Deep Learning**.

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5. Matrix and Derivative.

{Finding the Slope for Univariate Function.}

5.1 What is Derivative?

- The **derivative of a function** measures **how** the **output value** of **the function** changes as we make **small adjustments** to its **input**.
- Notations:
 - The derivative of a function $f(x)$ is represented by $\frac{d}{d(x)} (f(x))$ or $\frac{df(x)}{d(x)}$ or $f'(x)$ and is defined as:
- If we have a function $f(x)$, the derivative $f'(x)$ at a point x tell us the rate of change of function f at that point.
- This rate of change is crucial for **optimization techniques**,
 - such as **finding maxima or minima**, which are frequently used in **training machine learning models**.

5.2 Derivatives: Scalar Function.

- Most popular:
 - Derivative of a Scalar function i.e. Scalar derivatives $f: \mathbb{R} \rightarrow \mathbb{R}$
 - A scalar function is a function that maps a **real number x** to another **real number $f(x)$** .
 - $f(x) = x^2$
 - Here x : a **real number** and $f(x)$: also a real number.
 - We are interested in the **rate** at which **$f(x)$ changes** as **x changes**.
 - The derivative is the heart of calculus, buried inside this definition:
 - $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ when the limit exists.
 - popularly known as the “**limit definition of the derivative**” or “**derivative by using the first principle**”
 - But what does it mean?

5.2.1 Derivative First Principle: Interpretation.

- Derivative of a function is a measure of local slope.
 - 1st Example: For Linear Function $y = f(x) = 2x$.

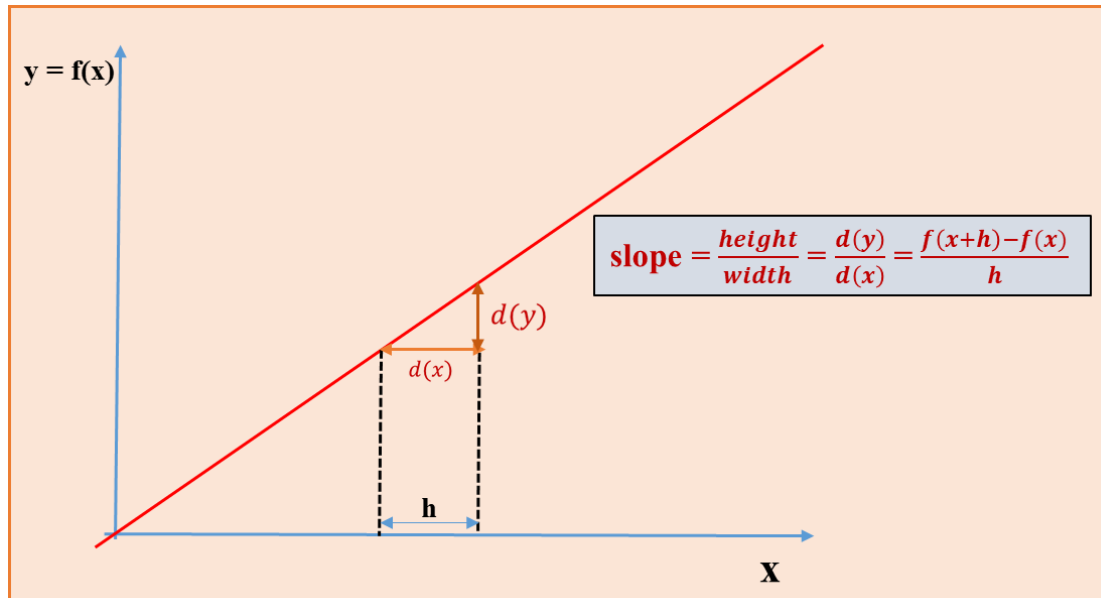
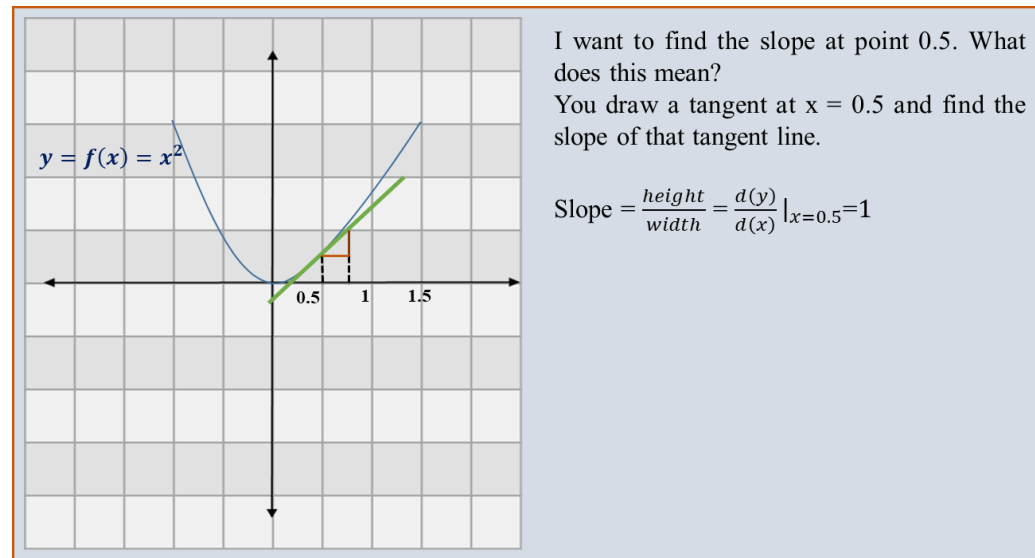


Fig: Derivative $\rightarrow$ Interpretation.

What for non linear function?

5.2.2 Derivative First Principle: Interpretation.

- 2nd Example: For Non - Linear Function $y = f(x) = x^2$.



- The derivative of a function at a **point is the slope of the tangent drawn to that curve** at that point.
 - (slope) derivative of a linear function (straight line) is constant at all the point not for the non-linear function.
- It also represents the **instantaneous rate of change** at a point on the function.

5.3 Some Common Rules for determining Derivative.

Rule	Function	Derivative
Sum – Difference Rule	$f(x) \pm g(x)$	$f'(x) \pm g'(x)$
Multiplication by Constant	$c \cdot f(x)$	$c \cdot f'(x)$
Product Rule	$f(x) \cdot g(x)$	$f'(x) \cdot g(x) + f(x) \cdot g'(x)$
Quotient Rule	$\frac{f(x)}{g(x)}$	$\frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$
Chain Rule	$f(g(x))$	$f'(g(x)) \cdot g'(x)$

!!! Hands on practice in Tutorial.

5.4 Derivative of some common function.

Function - Type	Function - Notation	Derivative
Constant function	$f(x) = c$; where c is real constant.	$f'(x) = (c)' = 0$.
Identity function	$f(x) = x$	$f'(x) = (x)' = 1$.
Linear function	$f(x) = mx$	$f'(x) = (mx)' = m$.
Function of the form	$f(x) = x^n$	$f'(x) = (x^n)' = nx^{n-1}$.
Exponential function of the form	$f(x) = a^x$; where $a > 0$	$f'(x) = (a^x)' = a^x \ln(a)$.
Exponential function	$f(x) = e^x$	$f'(x) = (e^x)' = e^x$.
Logarithmic function	$f(x) = \ln(x)$	$f'(x) = (\ln(x))' = \frac{1}{x}$.
Sinusoidal function	$f(x) = \sin(x)$	$f'(x) = (\sin(x))' = \cos(x)$.
Cosine function	$f(x) = \cos(x)$	$f'(x) = (\cos(x))' = -\sin(x)$.
Tangent function	$f(x) = \tan(x)$	$f'(x) = (\tan(x))' = \sec^2(x)$.

6. Matrix and Derivative.

{Finding the Slope for Multi – Variate Function.}

6.1 Derivative of a Multivariate Function.

- (**scalar derivative of**) Multivariate function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ are in the form $f(x, y) = x^2y$.
- **Partial Derivative:**
 - In mathematics, a **partial derivative** of a function of several variables is its derivative with respect to one of those variables, with the others held constant (as opposed to the total derivative, in which all variables are allowed to vary).
 - This swirly-d symbol, ∂ , often called "del", is used to distinguish partial derivatives from ordinary single-variable (regular) derivatives.
 - For Example: $f(x, y) = x^2y$.

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial x} x^2 y = y \frac{\partial}{\partial x} x^2 = 2xy$$

Treat y as a constant, then take regular derivative.

$$\frac{\partial}{\partial y} = \frac{\partial}{\partial y} x^2 y = x^2 \frac{\partial}{\partial y} y = x^2 \cdot 1$$

Treat x as a constant, then take regular derivative.

Derivative of $f(x, y) = x^2y$ are $2xy; x^2$

- Partial derivatives are used in vector calculus and differential geometry.

6.2 {some popular} Nomenclature of Derivative.

- Derivative of a vector/matrix a.k.a Matrix/Vector Calculus is an extension of ordinary scalar derivative to higher dimensional settings.
- Overview of some extended derivative style:

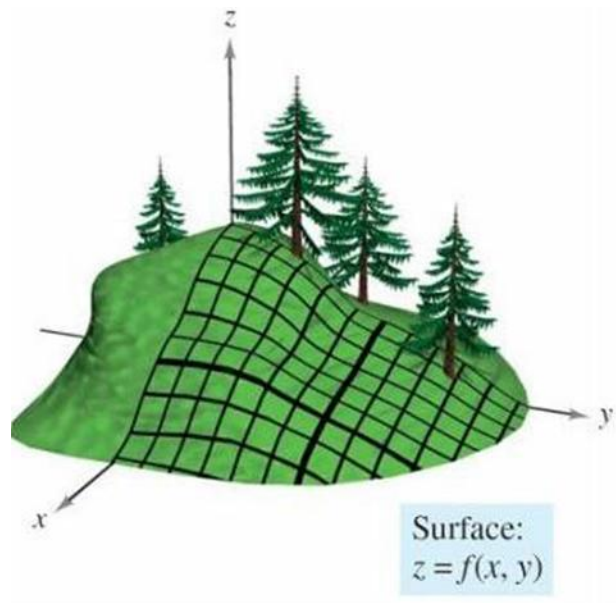
Setting	Derivative	Notation
$f: \mathbb{R} \rightarrow \mathbb{R}$	Scalar Derivative	$f'(x)$
$f: \mathbb{R}^n \rightarrow \mathbb{R}$	Gradient	$\nabla f(x)$
$f: \mathbb{R}^{n \times m} \rightarrow \mathbb{R}$	Gradient	$\nabla f(x)$
$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$	Jacobian	J_f

6.3 Gradient.

- Gradient:
 - The gradient of a function of multiple variables is the vector of partial derivatives of the function with respect to each variable.
 - Scalar-by-vector $\{f: \mathbb{R} \rightarrow \mathbb{R}^n\}$:
 - The derivative of a scalar function y with respect to a vector $x = [x_1, x_2, \dots, x_n]^T \in \mathbb{R}^n$ is written as:
 - gradients of y : $\nabla y = \frac{\partial y}{\partial x} = \left[\frac{\partial y}{\partial x_1} \quad \frac{\partial y}{\partial x_2} \quad \dots \quad \frac{\partial y}{\partial x_n} \right]^T$ — gradients.
 - {Stack the partial derivative against all the element of vector x }
 - Scalar-by-Matrix $\{f: \mathbb{R} \rightarrow \mathbb{R}^{n \times m}\}$:
 - The derivative of a scalar function y with respect to a $n \times m$ matrix X is written as:
 - gradients of y : $\nabla y = \frac{\partial y}{\partial X} = \begin{bmatrix} \frac{\partial y}{\partial x_{11}} & \dots & \frac{\partial y}{\partial x_{n1}} \\ \vdots & \ddots & \vdots \\ \frac{\partial y}{\partial x_{1m}} & \dots & \frac{\partial y}{\partial x_{nm}} \end{bmatrix}$
 - {Stack the partial derivative against all the element of Matrix X .}

**gradient is also the direction of steepest ascent,
What does that mean?**

6.4 Gradient: Geometric Interpretation.



He want's to scale the hill:

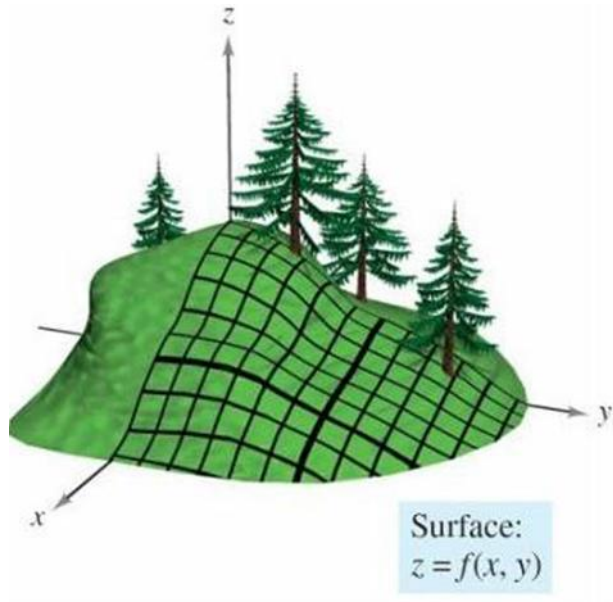
Let's assume he can take two routes:

One through $x \rightarrow$ co-ordinate(direction)

One through $y \rightarrow$ co-ordinate(direction)

Which route would be fastest?

6.4.1 Gradient: Geometric Interpretation.



He want's to scale the hill:

Let's assume he can take two routes:

One through $x \rightarrow$ co-ordinate(direction)

One through $y \rightarrow$ co-ordinate(direction)

Which route would be fastest?

Whichever direction has highest slope(gradient) i.e.

Find the gradient of the surface:

$$z = f(x, y)$$

gradient is a partial derivative of z against x and y stack in the vector.

This is read as: "grad. of z " or "grad z " $\leftarrow \nabla z = \left[\frac{\partial f(x, y)}{\partial x} \quad \frac{\partial f(x, y)}{\partial y} \right]$

6.5 Gradient: Example 1.

- $z = f(x, y) = 3x^2y$ find the gradient of z at $[1, 1]$.

- We know gradient of z is :

- $\nabla z = \left[\frac{\partial f(x, y)}{\partial x} \quad \frac{\partial f(x, y)}{\partial y} \right]$

- Finding: $\frac{\partial f(x, y)}{\partial x}$ i.e. y is constant.

$$\frac{\partial f(x, y)}{\partial x} = \frac{\partial 3yx^2}{\partial x} = \frac{3y \partial x^2}{\partial x} = 3y \cdot 2x = 6yx$$

- Finding $\frac{\partial f(x, y)}{\partial y}$ i.e. x is constant.

$$\frac{\partial f(x, y)}{\partial y} = \frac{\partial 3yx^2}{\partial y} = \frac{3x^2 \partial y}{\partial y} = 3x^2 \times 1 = 3x^2$$

- ∇z is:

$$\nabla z = [6yx \quad 3x^2]$$

- ∇z at $[1 \quad 1]$:

$$\nabla z = [6 \times 1 \times 1 \quad 3 \times 1^2] = [6 \quad 3]$$

6.6 Gradient of a Vector – Valued Function : Jacobian.

- **vector-by-vector** $\{f: \mathbb{R}^n \rightarrow \mathbb{R}^m\}$:

- The derivative of a vector function : $\mathbf{y} = [y_1, y_2, \dots, y_n]^T \in \mathbb{R}^n$ with respect to an input vector $\mathbf{x} = [x_1, x_2, \dots, x_m]^T \in \mathbb{R}^m$ is written as:

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} \\ \frac{\partial y_1}{\partial x_2} \\ \vdots \\ \frac{\partial y_n}{\partial x_1} \\ \frac{\partial y_n}{\partial x_2} \\ \vdots \\ \frac{\partial y_n}{\partial x_m} \end{bmatrix} = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \cdots & \frac{\partial y_1}{\partial x_m} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \cdots & \frac{\partial y_2}{\partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_n}{\partial x_1} & \frac{\partial y_n}{\partial x_2} & \cdots & \frac{\partial y_n}{\partial x_m} \end{bmatrix} = \mathbf{J}_y$$

- $\mathbf{J}_y$: called **Jacobian matrix** is a matrix **which contains all the partial derivatives** of each output component with respect to **each input variable**, providing a full picture how the vector-valued function changes as each input variable changes.

6.7 Derivative: Key Point

- The **derivative** of a **univariate function** is a **scalar**,
 - When the **derivative** of a **multivariate function** is organized and stored in a **vector**, the so-called **gradient**.
 - we denote the **derivative of a multivariable function f** using the **gradient symbol Δ** {read “del” or “nabla”}

$$\Delta f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \\ \vdots \end{bmatrix}$$

- The **gradient** is simply a **vector** listing the **derivatives of a function** with respect to **each argument of a function**.

The – End.